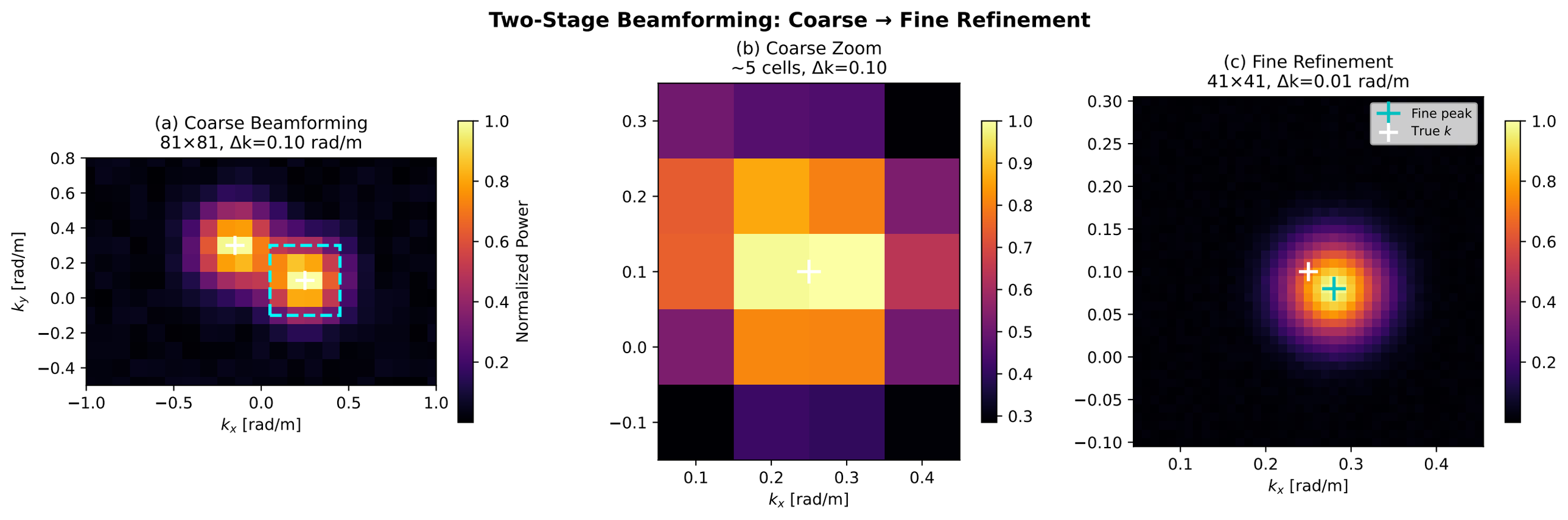

Two-Stage Beamforming: Coarse → Fine Refinement
(a) Coarse Beamforming
81×81, Δk=0.10 rad/m
k_y [rad/m]
k_x [rad/m]
Normalized Power
(b) Coarse Zoom
~5 cells, Δk=0.10
k_x [rad/m]
(c) Fine Refinement
41×41, Δk=0.01 rad/m
Fine peak
True k
k_x [rad/m]